Iterative Thresholding Framework for Row-Action Reconstruction from Sparse Projection Data

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***Abstract*—This paper presents a framework for row-action type iterative reconstruction from projection data measured over small number of projection views, or sparse projections. Image reconstruction from sparse projections usually suffers from streak artifacts that degrade the image quality. However, this imaging scenario becomes a hot topic of research due its possibilities to reduce patient dose and other benefits in several imaging applications. The motivation behind this work is the use of ℓ_1/ℓ_0 norm distance to a reference image to select the sparse solution corresponding to the difference between the reconstructed image and the reference image. The concept of Lagrangian duality is used to derive the iterative thresholding framework which can be classified as an ART-like reconstruction method. The work presented here can be thought as a generalization of the previous work in (Li et al., 2004) which mainly focuses on the reconstruction of sparse objects, such as blood vessels. However, most of clinical applications consider imaging of non-sparse objects. The extension to the general case of x-ray computed tomography (CT) imaging, where the target object is non-sparse, is presented here. Experimental data indicates the power of the proposed method in reducing streak artifacts produced from data down-sampling.**

I. Introduction

This paper presents a framework for row-action type iterative reconstruction from projection data measured over small number of projection views, or sparse projections. Image reconstruction from sparse projections usually suffers from streak artifacts that degrade the image quality. However, this imaging scenario becomes a hot topic of research due its possibilities to reduce patient dose and other benefits in several imaging applications [1]–[4]. The motivation behind this work is the use of ℓ_1/ℓ_0 norm distance to a reference image to select the sparse solution corresponding to the difference between the reconstructed image and the reference image. The concept of Lagrangian duality is used to derive the iterative thresholding framework which can be classified as an ART-like reconstruction method. The work presented here can be thought as a generalization of the previous work in [5] which mainly focuses on the reconstruction of sparse objects such as blood vessels. However, most of clinical applications consider imaging of non-sparse objects. The extension to the general case of x-ray computed tomography (CT) imaging, where the target object is non-sparse, is presented here.

This work was partially supported by Grant-in-Aid from the Japan Society for the Promotion of Science (JSPS). Grant for foreign post-doctoral fellows (ID No. P10052).

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II. Proposed Method

A. Problem formulation

In earlier work [5], the sparse solution to the linear constraint is picked up using the linear program:

$$\text{minimize } \|\vec{x}\|_1 \text{ subject to } \vec{x} \geq 0, A\vec{x} = \vec{y}, \tag{1}$$

where $\vec{x} = (x_1, \ldots, x_n)^\top$ is the image vector, $\vec{y} = (y_1, \ldots, y_m)^\top$ is the projection data and $A = \{a_{ij}\}$ is the system matrix. This cost function was formulated with consideration that $\vec{x}$ is sparse (i.e. the object to be imaged has only a small number of non-zero pixels). The motivation here is how to implement this method in general x-ray CT imaging, where the object is usually non-sparse. The formulation of sparse object is possible if some *a priori* information, such as a reference image, is available. The abnormalities, such as lesions and cancer indicators, usually appear in small regions of the imaged object. Therefore, subtracting the normal structures known in prior to the reconstruction is expected to result in a sparse image domain that includes unexpected or abnormal structures. This assumption holds if the registration error between the reference image and the reconstructed image is small. The proposed linear program to be considered here is as follows:

$$\text{minimize } \|\vec{x} - \vec{u}\|_1 \text{ s.t. } \vec{x} \geq 0, A\vec{x} = \vec{y}, \tag{2}$$

where $\vec{u} = (u_1, \ldots, u_n)^\top$ is a reference image to $\vec{x}$. The minimization of (2) is applied by computing the corresponding dual problem using Lagrangian duality. The maximization of the dual problem corresponds to the minimization of the original problem [6]. To approximate the solution, the linear program in Eq. (2) is perturbed to get the following quadratic program:

$$\text{minimize } \sum_{j=1}^{n} \left(\frac{\epsilon}{2}(x_j - u_j)^2 + |x_j - u_j| \right) \text{ s.t. } A\vec{x} = \vec{y}, \vec{x} \geq 0, \tag{3}$$

where ϵ is a small positive number. Similar to the implementation in [5], Eq. (3) is re-written as follows:

$$\text{minimize } F_{\vec{u}}(\vec{x}) \text{ s.t. } A\vec{x} = \vec{y}, \tag{4}$$

$$F_{\vec{u}}(\vec{x}) = \sum_{j=1}^{n} f_{u_j}(x_j), \quad f_a(t) = \frac{\epsilon}{2}(t-a)^2 + |t-a|.$$

The Lagrangian dual problem of (4) is the following unconstrained maximization problem:

$$\text{maximize } D(\vec{\lambda}) = \vec{y}^\top \vec{\lambda} - G_{\vec{u}}(A^\top \vec{\lambda}), \tag{5}$$

where $\vec{\lambda} = (\lambda_1, \ldots, \lambda_m)^\top$ is a real vector and $G_{\vec{u}}(.)$ is known as the conjugate function and is defined as:

$$G_{\vec{u}}(\vec{b}) = \sum_{i=1}^{n} g_{u_j}(b_j), \tag{6}$$

$$g_a(t) = \begin{cases} -\frac{1}{2}\epsilon a^2 - a & (t < -1 - \epsilon a) \\ at + \frac{(t+1)^2}{2\epsilon} & (-1 - \epsilon a \le t < -1) \\ at & (-1 \le t < 1) \\ at + \frac{(t-1)^2}{2\epsilon} & (t \ge 1) \end{cases}. \tag{7}$$

Assuming that $\vec{x}^*$ and $\vec{\lambda}^*$ are the optimal solutions for (4) and (5), respectively, then $\vec{x}^*$ can be computed from $\vec{\lambda}^*$ with the primal-dual relation defined as:

$$x_j^* = g'_{u_j}\left(A^\top \vec{\lambda}^*|_j\right), \tag{8}$$

$$g'_a(t) = \begin{cases} 0 & (t < -1 - \epsilon a) \\ a + \frac{t+1}{\epsilon} & (t < -1) \\ a & (-1 \le t < 1) \\ a + \frac{t-1}{\epsilon} & (t \ge 1) \end{cases}. \tag{9}$$

B. Iterative reconstruction framework

The solution of the maximization problem in (6) is done by the Newton-like coordinate ascent method. The final form of the iteration is summarized as follows:

Compute reference image $\vec{u} = (u_1, \ldots, u_n)$
Initialization $w^{(0)} = (0, \ldots, 0)$, $k \leftarrow 0$
repeat
  $\vec{w}^{(k,1)} = \vec{w}^{(k)}$
  for $i = 1 : m$ **do**
    $\vec{w}^{(k,i+1)} = \vec{w}^{(k,i)} + \frac{y_i - \sum_{j=1}^{n} a_{ij} g'_{u_j}(w_j^{(k,i)})}{\frac{1}{\epsilon}\sum_{j=1}^{n} a_{ij}^2} \vec{a}_i$
  end
  $\vec{w}^{(k+1)} = \vec{w}^{(k,m+1)}$
  $k \leftarrow k + 1$
until *stopping criterion*;
for $j = 1 : n$ **do**
  $x_j^* = g'_{u_j}(w_j^{(k+1)})$
end

The same approach can be implemented using ℓ_0 norm distance to formulate the cost function:

$$\text{minimize } \|\vec{x} - \vec{u}\|_0 \text{ s.t. } \vec{x} \ge 0, A\vec{x} = \vec{y}. \tag{10}$$

In this case, Eq. (7) can be derived as follows:

$$g_a(t) = \begin{cases} c & (t < t_c(a, \epsilon)) \\ at & (t_c(a, \epsilon) \le t < \sqrt{2\epsilon}) \\ \frac{t^2}{2\epsilon} + at - 1 & (t \ge \sqrt{2\epsilon}) \end{cases} \tag{11}$$

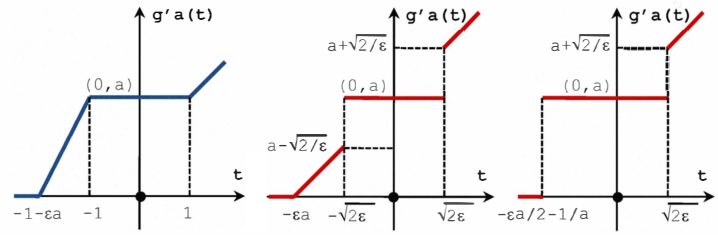

Fig. 1. Illustration of thresholding functions corresponding to Eq.(9) (left) and Eq.(13) with $a > \sqrt{2/\epsilon}$ (middle) and $a \le \sqrt{2/\epsilon}$ (right).

while $c = \min(\frac{t^2}{2\epsilon} + at - 1, -\frac{1}{2}\epsilon a^2 - 1)$ and

$$t_c(a, \epsilon) = \begin{cases} -\sqrt{2\epsilon} & (a > \sqrt{2/\epsilon}) \\ -\frac{1}{2}\epsilon a - \frac{1}{a} & (a \le \sqrt{2/\epsilon}) \end{cases}, \tag{12}$$

and the corresponding thresholding function, shown in Fig. 1, is given by:

$$g'_a(t) = \begin{cases} \max(a + \frac{t}{\epsilon}, 0) & (t < t_c(a, \epsilon)) \\ \max(a - \frac{1}{\sqrt{2\epsilon}}, \frac{a}{2}) & (t = t_c(a, \epsilon)) \\ a & (t_c(a, \epsilon) < t < \sqrt{2\epsilon}) \\ a + \frac{1}{\sqrt{2\epsilon}} & (t = \sqrt{2\epsilon}) \\ a + \frac{t}{\epsilon} & (t > \sqrt{2\epsilon}) \end{cases}. \tag{13}$$

The thresholding functions in Eqs. (9) and (13), shown in Fig. 1, have a close form to the soft/hard-thresholding operations used for image denoise in image processing literature. Unlike the case of ℓ_1 norm, the convergence of the ℓ_0 norm-based method is not assured.

III. Experimental Results

A simulation study was performed using a modified version of the Shepp-Logan phantom (Fig. 2). This study aims to observe the inserted spots with different reconstruction methods from small number of projection views. The image size was 512×512 pixels that is projected using 512(bins)×16(views) over 180° using parallel-beam geometry. The reference image $\vec{u}$ is assumed to be a uniform image shown in Fig. 2(b). The simulation is performed using 50 and 100 iterations using the ℓ_1, ℓ_0 norm-based methods, respectively. Reconstructed images are shown in Fig. 2.

IV. Conclusion

We presented iterative image reconstruction algorithm from small number of projection views. The main approach is based on minimizing the ℓ_1/ℓ_0 norm distance to *a priori* known reference image. Obviously, the accuracy of the reference image $\vec{u}$ has a strong contribution to the quality of the reconstructed image $\vec{x}$. However, we have found that a rough estimate, that could be obtained easily in several imaging application, is enough to eliminate streak artifacts significantly. The proposed row-action framework is simple, fast and easy to be implement to suit several imaging applications.

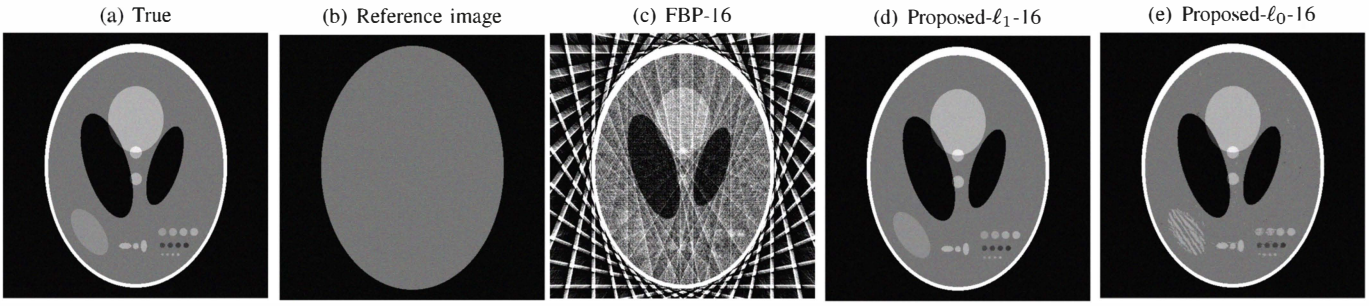

Fig. 2. (a) Modified Shepp-Logan phantom and image reconstructed from 16 projection views using (c) standard FBP algorithm and the proposed method using reference image in (b) with ℓ_1 and ℓ_0 norms in (d) and (e) , respectively.

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